DEEP CONVOLUTIONAL NEURAL NETWORKS FOR FINE-GRAINED SPICE IMAGE CLASSIFICATION

Course: CSE 414 – Machine Learning and Deep Learning Lab

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OUR CONTENTS

PROJECT OBJECTIVE

PROBLEM:

Spices are visually nearly identical, cumin vs. caraway, turmeric vs. mustard powder making automated identification extremely difficult even for humans. Existing food recognition systems either exclude spices entirely or group them into vague "seasoning" categories, making them impractical for real-world spice-specific tasks.

GOAL:

- Build SpiceFusionNet, a multi-modal deep learning model that classifies 11 spice types by fusing image, texture, and color features for robust fine-grained recognition.
- Achieve high accuracy and real-time inference speed, making it deployable in smart kitchen assistants, food quality control, and retail inventory management.

SOCIETAL IMPACT AND MOTIVATION

Why This Matters :

- Spice adulteration is a global food safety concern, automated identification helps detect mislabeling and contamination in food supply chains.
- Smart kitchen systems are increasingly common; accurate spice recognition enables hands-free cooking assistance and dietary tracking.
- Retail and e-commerce platforms can use automated spice recognition for inventory management and quality grading.

Who Benefits :

- Consumers: healthier, safer food choices through better labeling.
- Food industry: automated quality control and fraud detection.
- Retailers: faster, more accurate inventory categorization.
- Researchers: a specialized spice dataset and benchmark for future food-AI work.

Uniqueness: Unlike general food recognition models, this system's purpose-built for spices, addressing a clear research gap with a custom dataset, spice-specific augmentation, and contrastive training.

DATASET DESCRIPTION

Primary Dataset:
Custom Spice Image Dataset

- 11,000 images across 11 spice classes
- Sources: Open Images, iNaturalist, and self-captured images
- Each class: ~1,000 images in varied lighting, backgrounds, and forms (whole, ground, dried)
- Dataset reference: [ScienceDirect link]

Supplementary Datasets

- SPICE-30 Benchmark (Kaggle) for baseline comparison
- Food-101 spice subset for transfer learning pre-training
- Custom macro-photography subset (~2,000 images) for texture analysis

Data Split: Train 70% | Validation 10% | Test 20% (stratified)

Preprocessing:

- Resize to 512×512, normalize with ImageNet mean/std
- Background removal using SAM (Segment Anything Model)
- Augmentation: horizontal/vertical flip, rotation ±30°, color jitter, Gaussian blur

METHODOLOGY / MODEL ARCHITECTURE

Model: SpiceFusionNet :Three-Branch Multi-Modal Fusion Network

BRANCH	Input	Architecture	Output
CNN Backbone	Raw image (224×224×3)	EfficientNet-B4 (pretrained, timm) → Avg Pool	1792-d
Texture Branch	LBP (10-d) + GLCM (48-d) = 58-dexture patterns	MLP: 58→128→256 + BN + ReLU + Dropout(0.3)	256-d
Color Branch	HSV Histogram: H(36)+S(32)+V(32) = 100-d	MLP: 100→64→128 + BN + ReLU + Dropout(0.3)	128-d

METHODOLOGY / MODEL ARCHITECTURE

Fusion: AttentionFusion Module

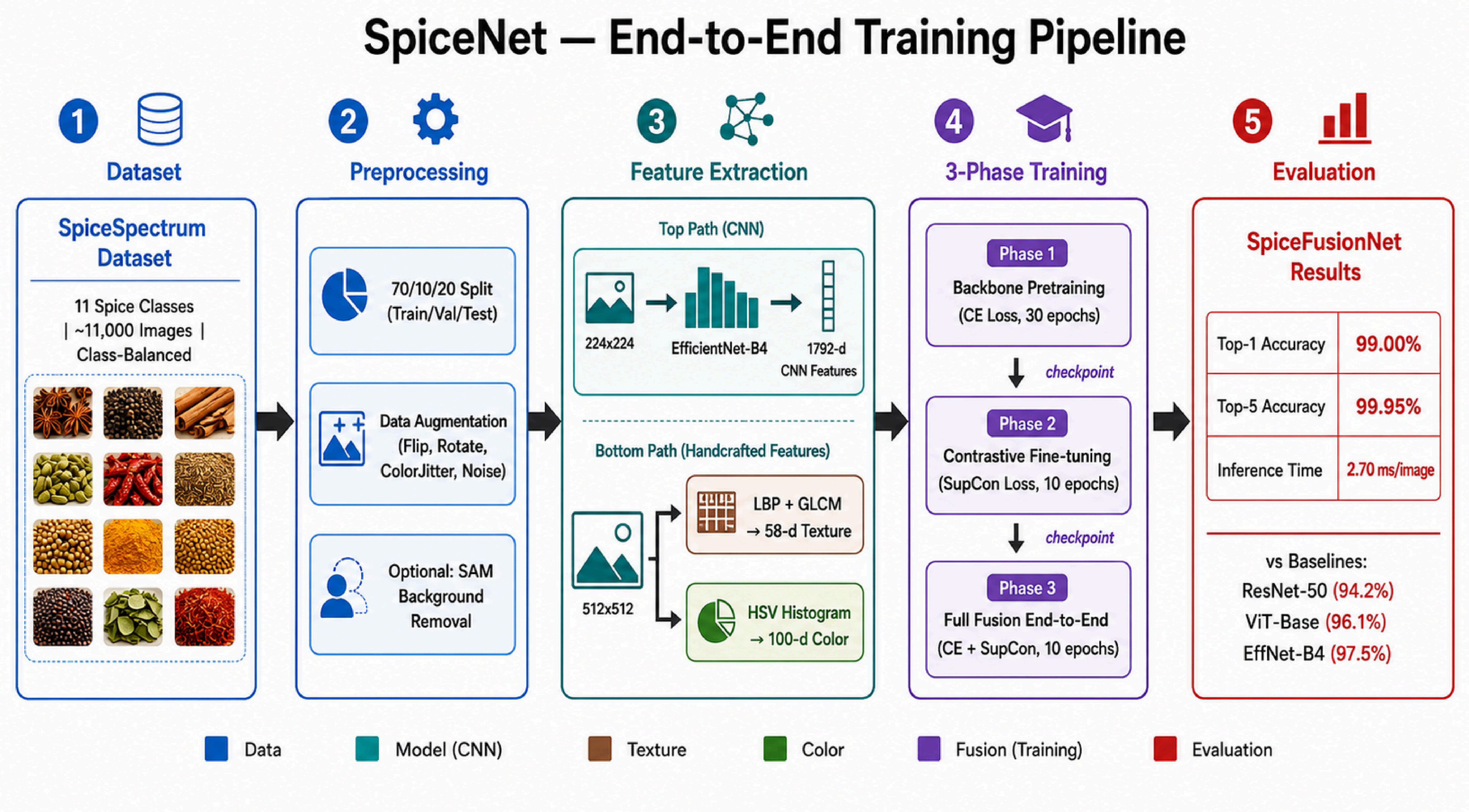
- Concatenates all three branch outputs → 2176-d (1792 + 256 + 128)
- Learns per-branch importance weights via a softmax gate (α_img, α_tex, α_col)
- Fused vector passed to fusion_head: MLP(2176→512→11)

Model Size:
21.6M parameters total
EfficientNet-B4 backbone: ~19.3M
Texture + Color MLPs: ~56K
Fusion head + attention: ~1.2M

Three-Phase Training Strategy:

Phase	What Trains	Loss	Epochs
Phase 1 – Supervised Pre-training	EfficientNet-B4 + img_head	Cross-Entropy + Label Smoothing	30
Phase 2 – Contrastive Fine-tuning	proj_head on hard-negative pairs (cumin/caraway, turmeric/mustard)	SupCon Loss	10
Phase 3 – Full Fusion Training	All branches + AttentionFusion + fusion_head	CE + SupCon	10

METHODOLOGY / MODEL ARCHITECTURE



IMPLEMENTATION

Tools & Frameworks:

Category	Tools Used
Language	Python 3.10
Classical Features	Classical Features
Augmentation	Albumentations
Interpretability	Grad-CAM
Experiment Tracking	Weights & Biases
Model Sharing	HuggingFace Hub

RESULTS AND ANALYSIS

Evaluation Metrics Used:

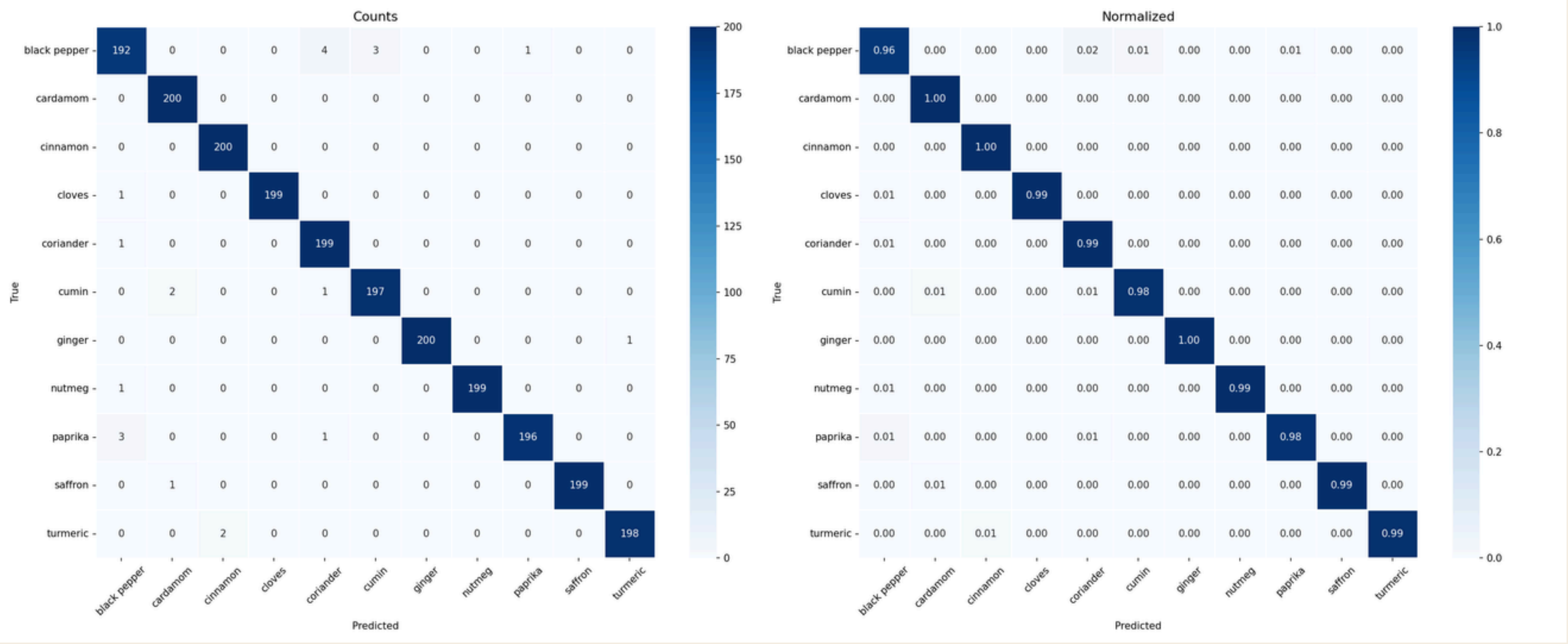
- Top-1 & Top-5 Classification Accuracy
- Macro-averaged F1 Score
- Per-class Precision & Recall
- Confusion Matrix (focus on hard-negative pairs)
- Inference time (ms/image)

BaseLine Comparisons:

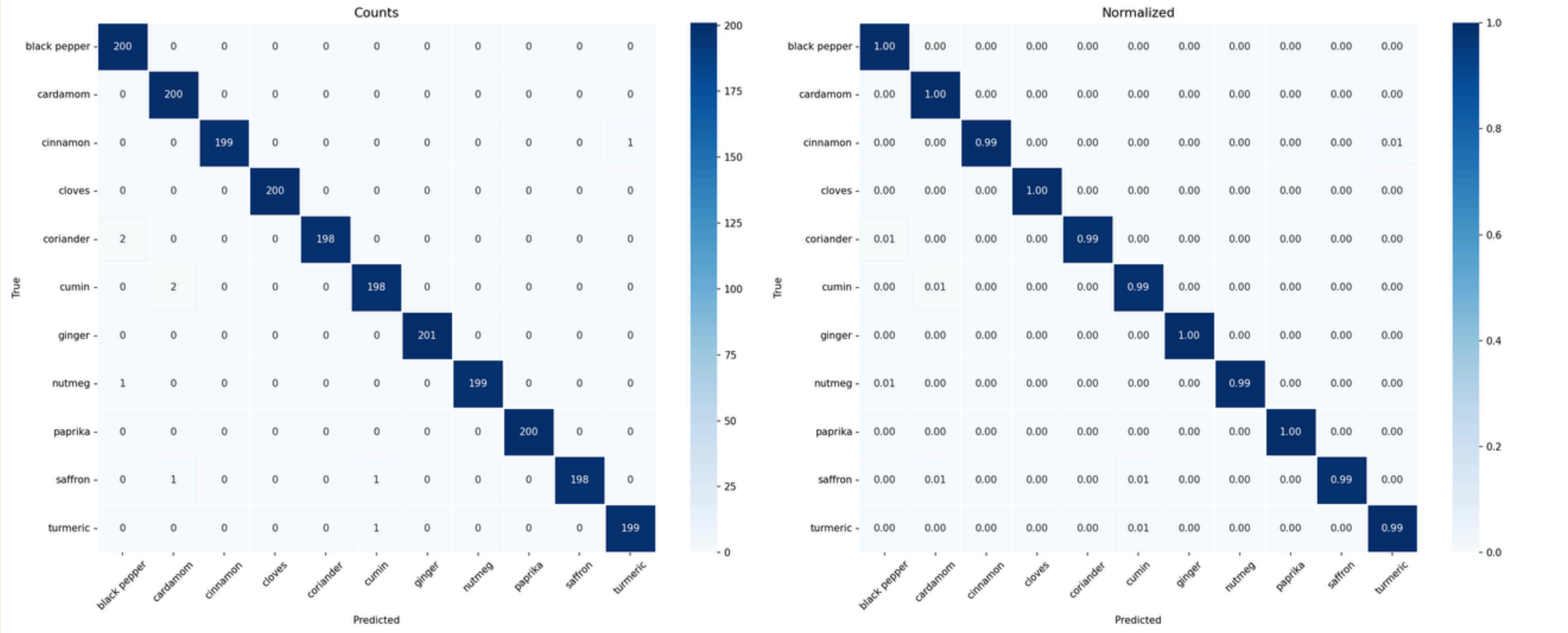
Model	Top-1 Accuracy	F1 Score
SVM (HOG + Color Histogram)	32.49	30.69
ResNet-50 (fine-tuned)	99.36	99.36
EfficientNet-B4 (image only)	99.59	99.59
ViT-Base/16 (fine-tuned)	99.73	99.73
SpiceFusionNet (ours)	99.68	99.68

CONFUSION MATRIX

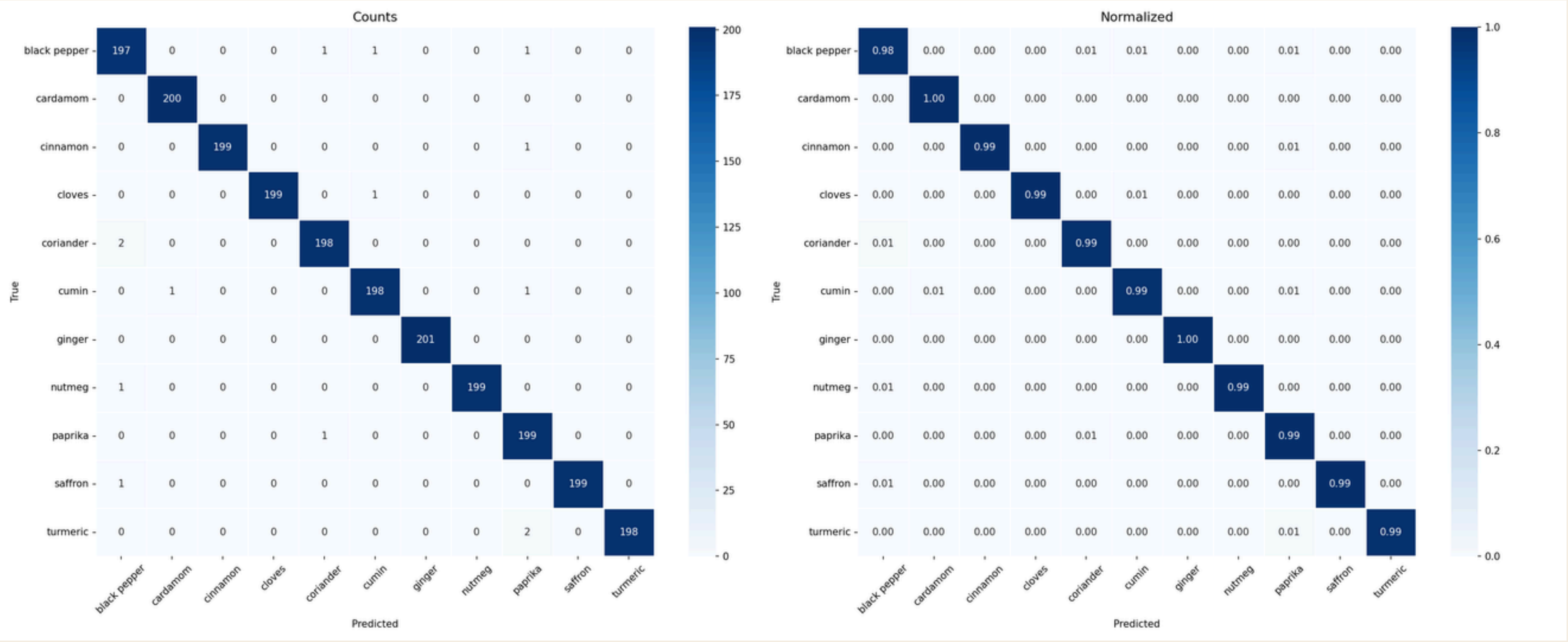
Fusion



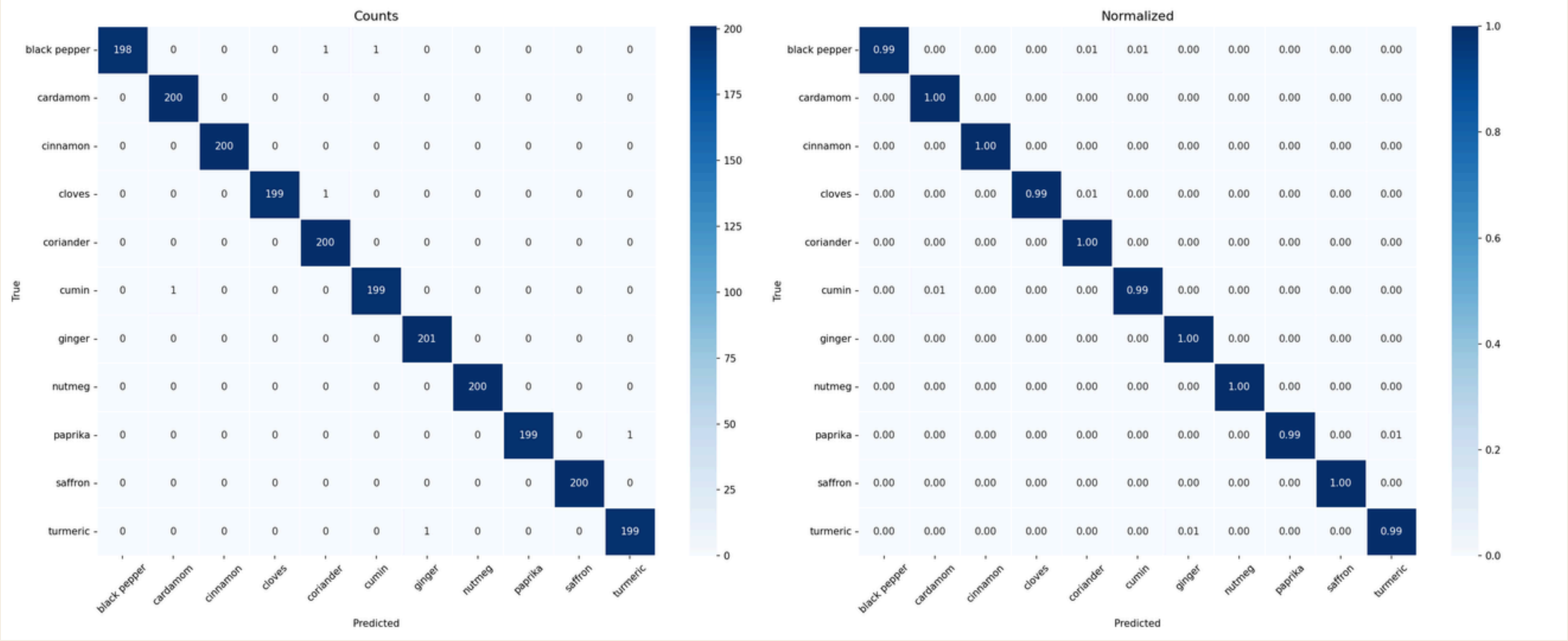
efficientnet_b4



Restnet50

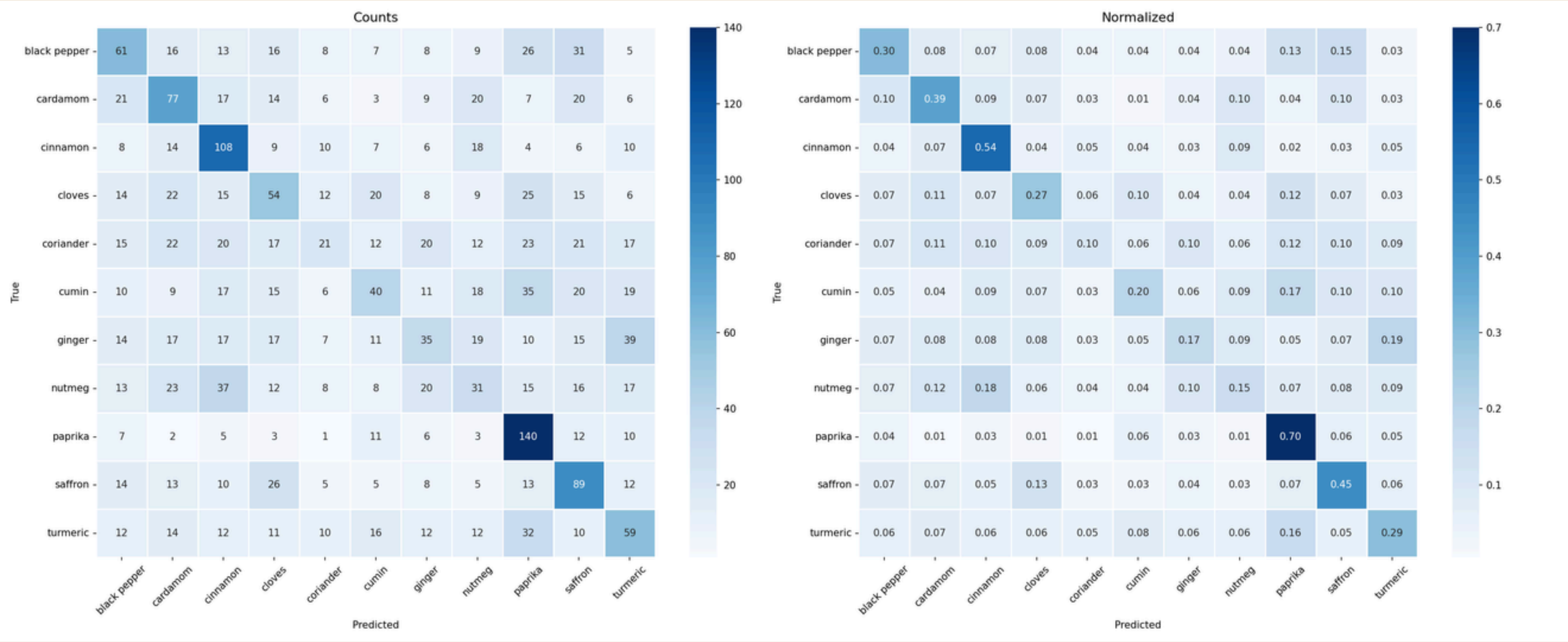


Vit_base

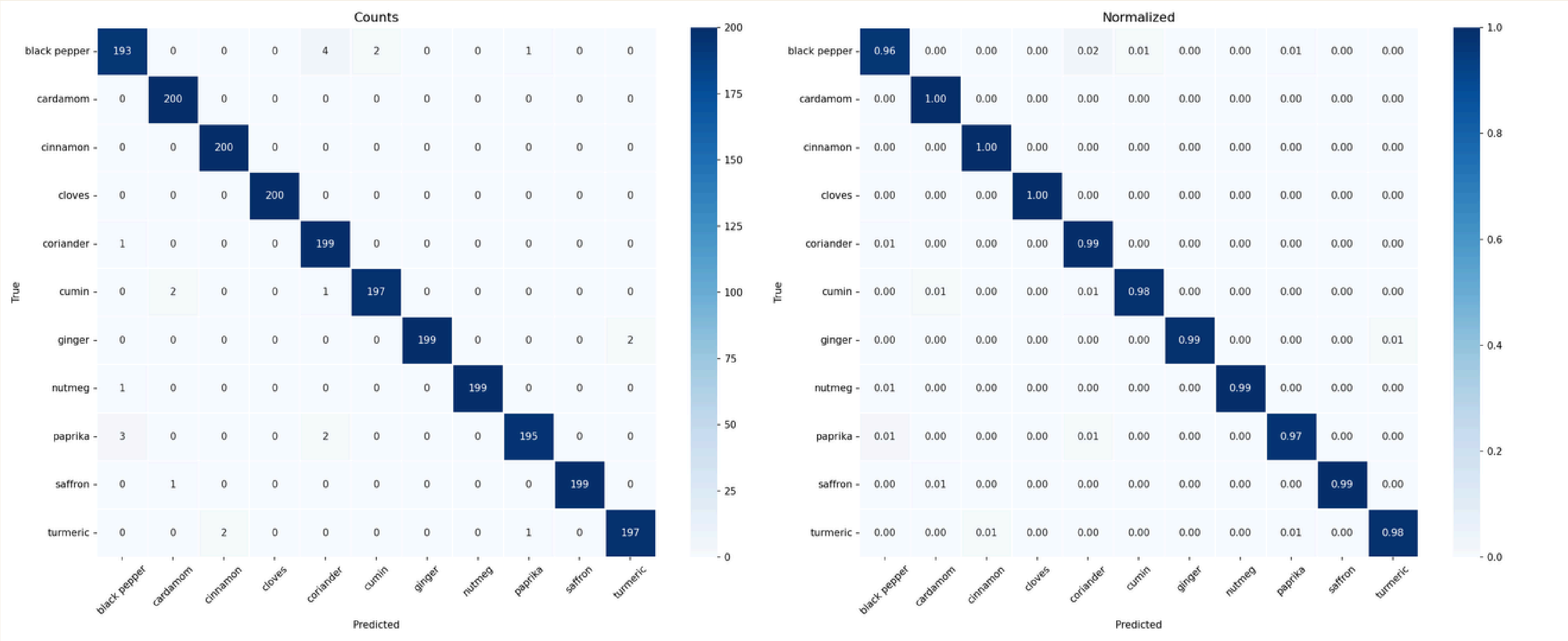


CONFUSION MATRIX

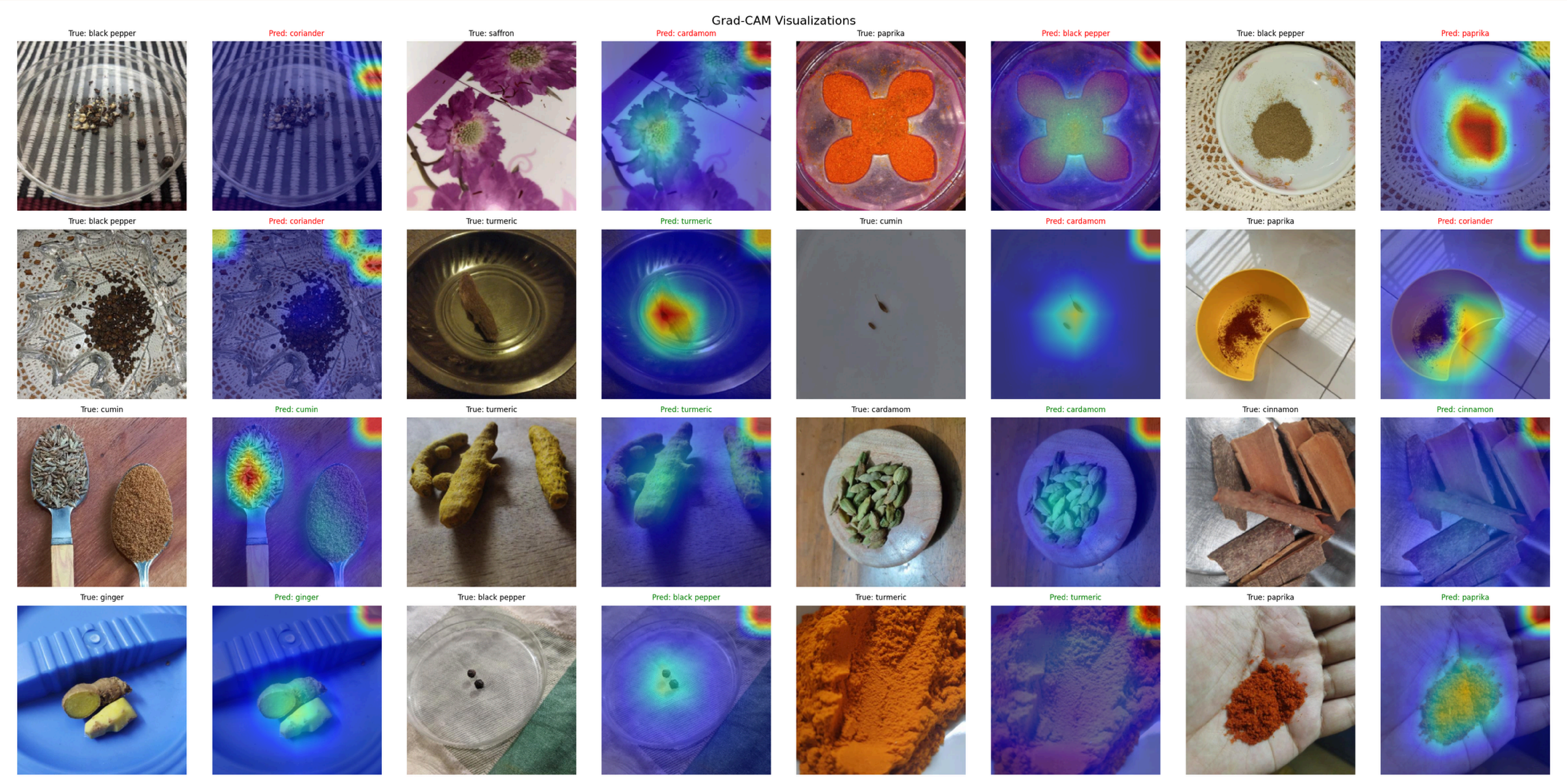
SVM



Image

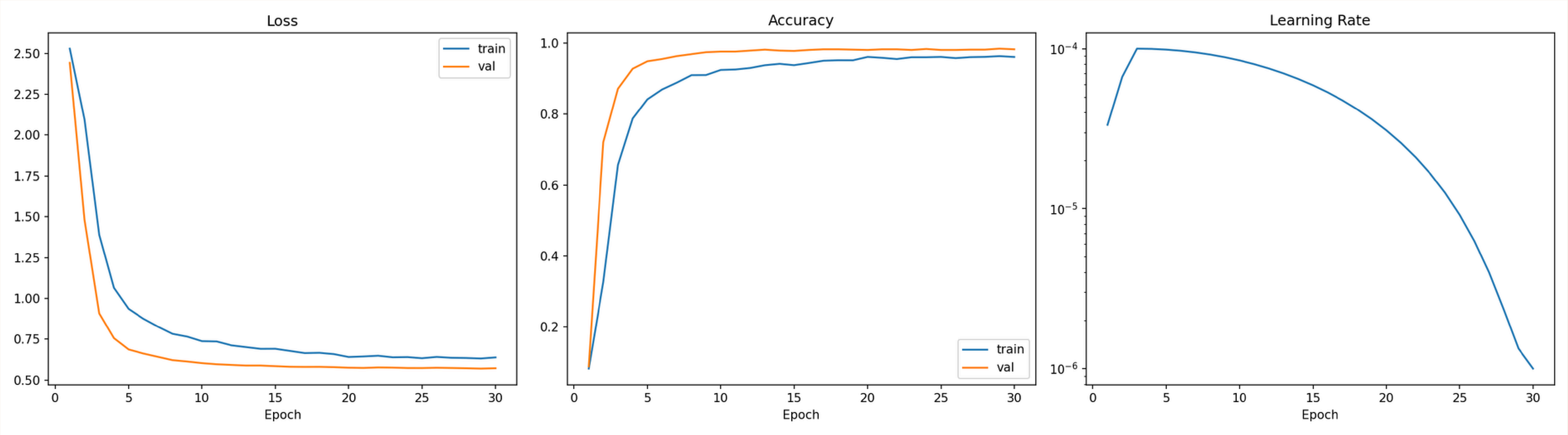


GRADCAM VISUALIZATION

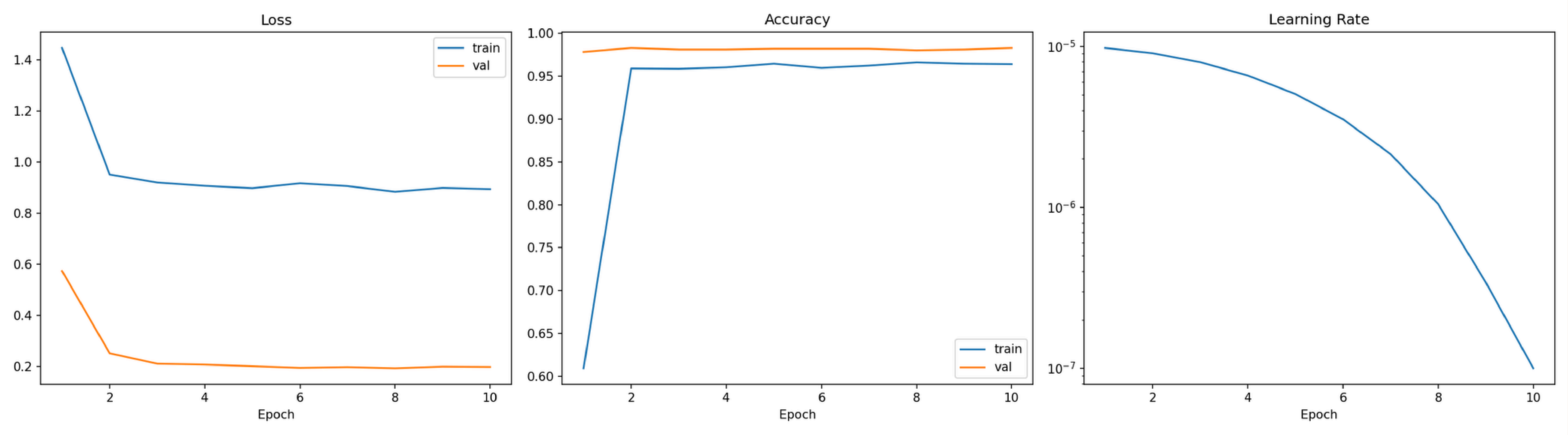


TRAINING LOSE, ACCURACY & LEARNING RATE CURVES

P1 training curves



P3 training curves



TEAM MEMBER CONTRIBUTIONS

Member	ID	Contribution
Md. Noushad Jahan Ramim	22201257	Model architecture design, EfficientNet-B4 backbone integration, Phase 1 & 2 training, report writing
Maisha Sameha	22201266	Dataset preprocessing, augmentation pipeline, SAM variant creation, report writing
Samira Islam	22201262	Texture & color branch implementation, attention fusion module, ablation studies, report writing
Junaid Abedin Rafi	22201265	Baseline model evaluation, GradCAM visualization, results analysis, presentation preparation

CONCLUSION

Summary:

Developed SpiceFusionNet, a multi-modal CNN that fuses image, texture, and color features to classify 11 spice types with high accuracy.

Contrastive fine-tuning on hard-negative pairs significantly reduced confusion between visually similar spices.

The system is suitable for real-time deployment in smart kitchen, food quality control, and retail use cases.

Challenges:

High visual similarity between spice classes made fine-grained discrimination difficult.

Limited availability of labeled spice datasets required significant custom data collection.

Background clutter impacted model performance; addressed partially with SAM-based segmentation.

Future Work:

Scale dataset to 30+ spice categories (SPICE-30 benchmark)

Deploy as a mobile app with on-device inference (MobileNetV3 lightweight variant)

Extend to multi-spice blend recognition and adulteration detection

REFERENCES

[1] GRAND VIEW RESEARCH, "SPICES MARKET SIZE, SHARE & TRENDS ANALYSIS REPORT," MARKET RESEARCH REPORT, 2023.

[2] SPICESPECTRUM DATASET, "CLASS-BALANCED DATASET OF COMMERCIALLY VALUABLE SPICE CULTIVARS," DATA IN BRIEF, ELSEVIER, 2025.

[3] L. BOSSARD, M. GUILLAUMIN, AND L. VAN GOOL, "FOOD-101 – MINING DISCRIMINATIVE COMPONENTS WITH RANDOM FORESTS," PROC. ECCV, 2014.

[4] K. HE, X. ZHANG, S. REN, AND J. SUN, "DEEP RESIDUAL LEARNING FOR IMAGE RECOGNITION," PROC. CVPR, 2016.

[5] M. TAN AND Q. V. LE, "EFFICIENTNET: RETHINKING MODEL SCALING FOR CONVOLUTIONAL NEURAL NETWORKS," PROC. ICML, 2019.

[6] P. KHOSLA ET AL., "SUPERVISED CONTRASTIVE LEARNING," PROC. NEURIPS, 2020.

[7] R. R. SELVARAJU ET AL., "GRAD-CAM: VISUAL EXPLANATIONS FROM DEEP NETWORKS VIA GRADIENT-BASED LOCALIZATION," PROC. ICCV, 2017.

[8] A. KIRILLOV ET AL., "SEGMENT ANYTHING," PROC. ICCV, 2023.

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THANK YOU

Any Questions?